

Dimitrios Panagiotis Emmanouilidis

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Software developer focused on local LLM inference, RAG, and multi-agent orchestration, with hands-on MLOps and edge-hardware deployment.

TECHNICAL SKILLS

Languages & OS	Python, Linux
Infrastructure & MLOps	Docker, Git, hardware optimization (eGPU / VRAM allocation), CI/CD concepts
AI & Machine Learning	LangGraph, Ollama, Retrieval-Augmented Generation (RAG), local LLM inference, multi-agent systems, sensor fusion

TECHNICAL PROJECTS

Offline Local Inference & RAG Application	<i>Lead Developer · In Development</i>
- Engineering a local LLM application prioritising off-grid knowledge retrieval without reliance on external cloud APIs. - Integrating heavily quantised local models optimised for consumer hardware, managing execution across mobile and eGPU environments. - Architecting a built-in Retrieval-Augmented Generation (RAG) pipeline to dynamically query offline databases (e.g. Wikipedia) for real-time factual generation.	
Autonomous Vehicle Multi-Agent Simulator	<i>System Architect · Early 2026</i>
- Architected a multi-agent AI orchestration pipeline using LangGraph for an autonomous vehicle simulation environment. - Developed and integrated real-time sensor fusion algorithms to process simulated environmental data streams. - Optimised system decision-making by reducing model latency under strict real-time execution constraints.	
Local AI Infrastructure & Orchestration Deployment	<i>DevOps & Hardware Ops · Late 2025 – Present</i>
- Configured and deployed a high-performance local AI execution server using a Beelink GTi13 Ultra and an external NVIDIA RTX 5070 GPU. - Containerised AI orchestration frameworks (LangGraph, Ollama) with Docker for reproducible, isolated environments. - Optimised local inference performance through aggressive VRAM management and secure hardware bridging.	
Curated Map-Based Travel Application	<i>Backend Developer · June 2026 – Present</i>
- Developing the backend architecture for a mobile travel application focused on memory curation and itinerary mapping. - Designing real-time data-fetching logic and database schemas to support a map-based “storybook” recap system and custom data-visualisation dashboards.	

CERTIFICATIONS

- AWS Certified Generative AI Developer (AIP-C01) — Expected Jan 2027
- AWS Certified Machine Learning Engineer (MLA-C01) — Expected Oct 2026
- MLOps Zoomcamp (DataTalks.Club) — Expected Sep 2026
- DeepLearning.AI Machine Learning Specialization — Expected Aug 2026
- Python Development Certification (LearnX) — Completed March 2025

EDUCATION

B.Sc. in Software Development	<i>Expected July 2026</i>
New York College / University of Greater Manchester — Thessaloniki, Greece	